Geometry & Trigonometry

Section 1: Angle Relationships & Polygons

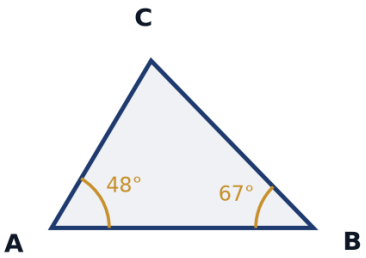
1 In triangle ABC, the measure of angle A is 48° and the measure of angle B is 72°. What is the measure of the exterior angle at vertex C?

a. 110°

b. 120°

c. 130°

d. 140°



2 What is the sum of the interior angle measures of a regular octagon (8-sided polygon)?

a. 900°

b. $1,080^\circ$

c. $1,260^\circ$

d. $1,440^\circ$

3 Two parallel lines are cut by a transversal. If one interior angle measures $(3x + 15)^\circ$ and its consecutive interior angle measures $(2x + 25)^\circ$, what is the value of x?

a. 24

b. 28

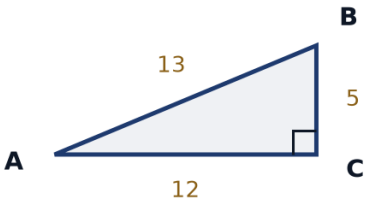
c. 32

d. 36

Section 2: Right Triangle Trigonometry & Special Triangles

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4 In right triangle ABC with right angle at C, $\sin A = \frac{5}{13}$. What is the value of $\cos B$?



a. $\frac{5}{13}$

b. $\frac{12}{13}$

c. $\frac{5}{12}$

d. $\frac{13}{5}$

5 If $\sin(2x - 10^\circ) = \cos(3x + 20^\circ)$, where all angles are acute, what is the value of x?

a. 14

b. 16

c. 18

d. 20

6 A $30^\circ - 60^\circ - 90^\circ$ right triangle has a shorter leg of length 7. What is the area of the triangle?

a. $\frac{49\sqrt{3}}{2}$

b. $49\sqrt{3}$

c. 24.5

d. 98

Section 3: Circles & Standard Equations

7 What is the radius of the circle with equation $x^2 + y^2 - 6x + 8y - 11 = 0$?

a. 5

b. 6

c. 7

d. 8

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8 A circle in the xy-plane has center $(4, -2)$ and passes through the point $(7, 2)$. What is the equation of the circle?

a. $(x-4)^2 + (y+2)^2 = 25$

b. $(x+4)^2 + (y-2)^2 = 25$

c. $(x-4)^2 + (y+2)^2 = 5$

d. $(x-7)^2 + (y-2)^2 = 25$

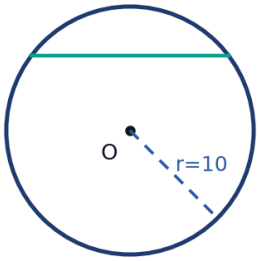
9 In a circle of radius 10, a chord is drawn at a distance of 6 units from the center. What is the length of the chord?

a. 8

b. 12

c. 16

d. 18



Section 4: Arc Lengths, Sector Areas & Volume

10 A circle has a radius of 12 cm. What is the length of an arc intercepted by a central angle of $\frac{2\pi}{3}$ radians?

a. 4π cm

b. 8π cm

c. 12π cm

d. 16π cm

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11 What is the area of a sector of a circle with radius 6 cm and central angle 60°?

a. 3π cm^2

b. 6π cm^2

c. 12π cm^2

d. 18π cm^2

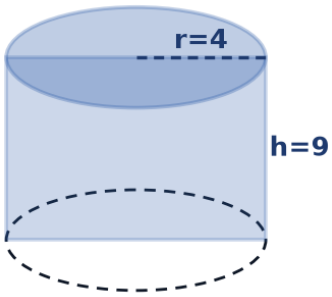
12 A right circular cylinder has a radius of 4 cm and a height of 9 cm. What is the volume of the cylinder?

a. 72π cm^3

b. 108π cm^3

c. 144π cm^3

d. 288π cm^3



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